

Conversion Function

The screenshot shows a SQL IDE interface with multiple tabs at the top labeled 'SQL File 10*' through 'SQL File 18*'. The active tab is 'SQL File 18*'. Below the tabs is a toolbar with various icons, including a 'Limit to 1000 rows' dropdown. The main area displays a SQL script with 15 lines of code. The script creates a database 'A', uses it, creates a table 'Student' with columns 'Student_ID', 'Student_Name', 'Marks', and 'DOB', inserts four rows of data, and then selects the 'Student_Name' and 'Marks' columns, casting 'Marks' to a character string.

```
1 • CREATE DATABASE A;  
2 • USE A;  
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Student_Name VARCHAR(50),  
6     Marks INT,  
7     DOB DATE  
8 );  
9 • INSERT INTO Student VALUES  
10 (101, 'Manu', 90, '2005-05-10'),  
11 (102, 'Varun', 80, '2004-08-15'),  
12 (103, 'Vaibhav', 70, '2005-01-20'),  
13 (104, 'Raghavendra', 60, '2004-12-05');  
14 • SELECT Student_Name, CAST(Marks AS CHAR) AS Marks_Text  
15 FROM Student;
```

Below the script, the 'Result Grid' is visible, showing the output of the query. It has columns 'Student_Name' and 'Marks_Text'. The results are:

Student_Name	Marks_Text
Manu	90
Varun	80
Vaibhav	70
Raghavendra	60

SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14* SQL File 16* SQL File 17* SQL File 18* x

Limit to 1000 rows

```
1 • CREATE DATABASE B;  
2 • USE B;  
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Student_Name VARCHAR(50),  
6     Marks INT,  
7     DOB DATE  
8 );  
9 • INSERT INTO Student VALUES  
10 (101, 'Manu', 90, '2005-05-10'),  
11 (102, 'Varun', 80, '2004-08-15'),  
12 (103, 'Vaibhav', 70, '2005-01-20'),  
13 (104, 'Raghavendra', 60, '2004-12-05');  
14 • SELECT Student_Name, CONVERT(Marks, CHAR) AS Marks_Text  
15 FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Student_Name	Marks_Text
▶	Manu	90
	Varun	80
	Vaibhav	70
	Raghavendra	60

Date Function

```

1 • CREATE DATABASE C;
2 • USE C;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Student_Name VARCHAR(50),
6     Marks INT,
7     DOB DATE
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 90, '2005-05-10'),
11 (102, 'Varun', 80, '2004-08-15'),
12 (103, 'Vaibhav', 70, '2005-01-20'),
13 (104, 'Raghavendra', 60, '2004-12-05');
14 • SELECT CURRENT_DATE();
15 • SELECT Student_Name, YEAR(DOB) AS Birth_Year
16 FROM Student;
17 • SELECT Student_Name, MONTH(DOB) AS Birth_Month

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	CURRENT_DATE()
►	2026-10-02

SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14* SQL File 16* SQL File 17* SQL File 18*

Limit to 1000 rows

```
1 • CREATE DATABASE C;
2 • USE C;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Student_Name VARCHAR(50),
6     Marks INT,
7     DOB DATE
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 90, '2005-05-10'),
11 (102, 'Varun', 80, '2004-08-15'),
12 (103, 'Vaibhav', 70, '2005-01-20'),
13 (104, 'Raghavendra', 60, '2004-12-05');
14 • SELECT CURRENT_DATE();
15 • SELECT Student_Name, YEAR(DOB) AS Birth_Year
16 FROM Student;
17 • SELECT Student_Name, MONTH(DOB) AS Birth_Month
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Student_Name	Birth_Year
▶	Manu	2005
	Varun	2004
	Vaibhav	2005
	Raghavendra	2004

SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14* SQL File 16* SQL File 17* SQL File 18* x

Limit to 1000 rows

```
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Student_Name VARCHAR(50),  
6     Marks INT,  
7     DOB DATE  
8 );  
9 • INSERT INTO Student VALUES  
10 (101, 'Manu', 90, '2005-05-10'),  
11 (102, 'Varun', 80, '2004-08-15'),  
12 (103, 'Vaibhav', 70, '2005-01-20'),  
13 (104, 'Raghavendra', 60, '2004-12-05');  
14 • SELECT CURRENT_DATE();  
15 • SELECT Student_Name, YEAR(DOB) AS Birth_Year  
16 FROM Student;  
17 • SELECT Student_Name, MONTH(DOB) AS Birth_Month  
18 FROM Student;  
19 • SELECT Student_Name, DAY(DOB) AS Birth_Day  
20 FROM Student;  
21 • SELECT Student_Name, DATEDIFF(CURRENT_DATE(), DOB) AS B
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

	Student_Name	Birth_Day
▶	Manu	10
	Varun	15
	Vaibhav	20
	Raghavendra	5

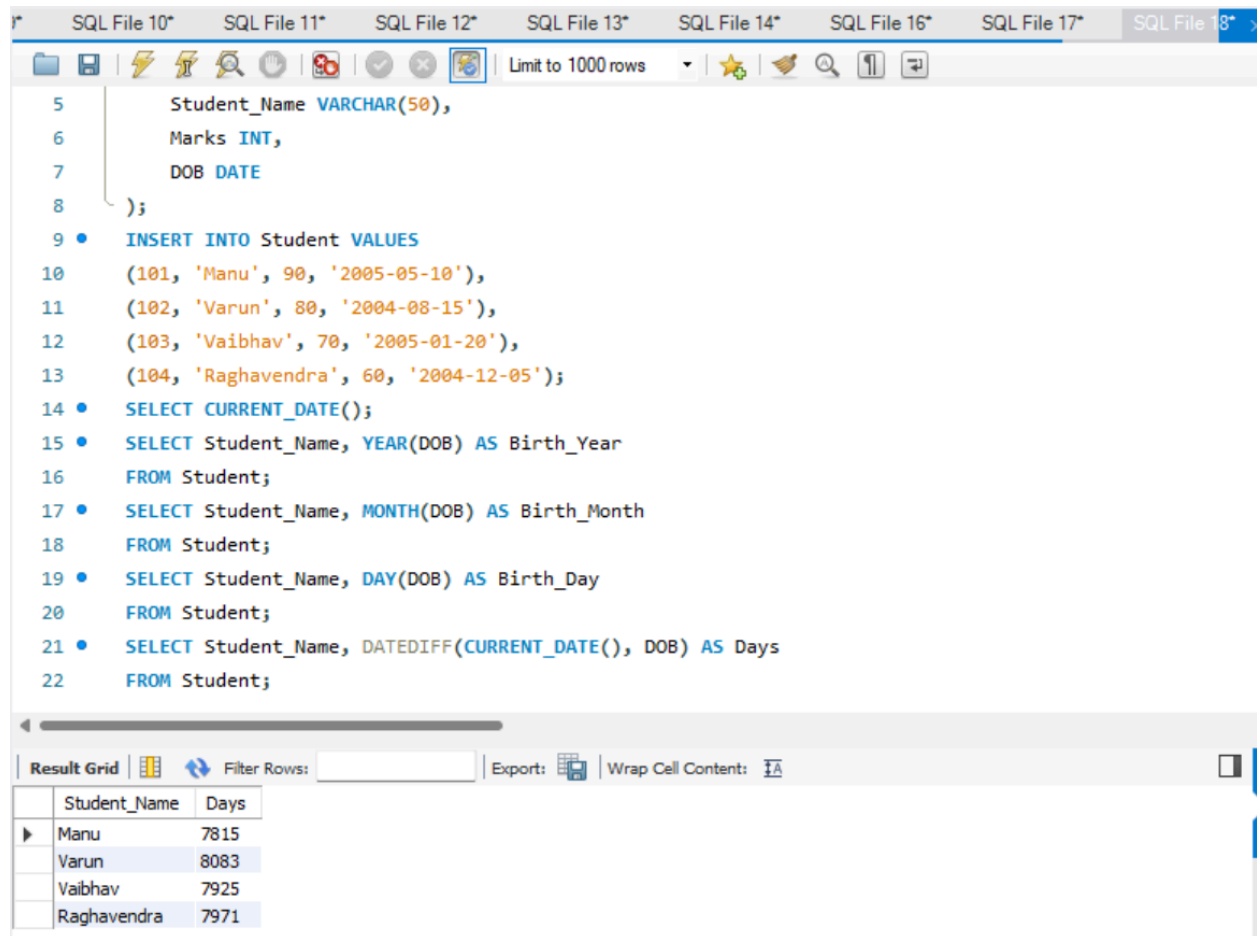
SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14* SQL File 16* SQL File 17* SQL File 18* x

Limit to 1000 rows

```
5      Student_Name VARCHAR(50),
6      Marks INT,
7      DOB DATE
8  );
9  • INSERT INTO Student VALUES
10 (101, 'Manu', 90, '2005-05-10'),
11 (102, 'Varun', 80, '2004-08-15'),
12 (103, 'Vaibhav', 70, '2005-01-20'),
13 (104, 'Raghavendra', 60, '2004-12-05');
14 • SELECT CURRENT_DATE();
15 • SELECT Student_Name, YEAR(DOB) AS Birth_Year
16 FROM Student;
17 • SELECT Student_Name, MONTH(DOB) AS Birth_Month
18 FROM Student;
19 • SELECT Student_Name, DAY(DOB) AS Birth_Day
20 FROM Student;
21 • SELECT Student_Name, DATEDIFF(CURRENT_DATE(), DOB) AS Days
22 FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Student_Name	Birth_Day
▶	Manu	10
	Varun	15
	Vaibhav	20
	Raghavendra	5



SQL Operators – Arithmetic, Logical and Comparison

```

CREATE DATABASE D;
USE D;
CREATE TABLE Student (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Marks INT,
    DOB DATE
);
INSERT INTO Student VALUES
(101, 'Manu', 90, '2005-05-10'),
(102, 'Varun', 80, '2004-08-15'),
(103, 'Vaibhav', 70, '2005-01-20'),
(104, 'Raghavendra', 60, '2004-12-05');
SELECT Marks + 5 AS New_Marks
FROM Student;

```

```
SELECT Marks - 5 AS Reduced_Marks
FROM Student;
SELECT Marks * 2 AS Double_Marks
FROM Student;
SELECT Marks / 2 AS Half_Marks
FROM Student;
SELECT Marks % 10 AS Remainder
FROM Student;
```

Output

Result Grid		Filter Rows:	Export:	Wrap C
	New_Marks			
▶	95			
	85			
	75			
	65			

Result 8 × Result 9 Result 10 Result 11 Result 12

Result Grid		Filter Rows:	Export:	Wrap C
	Reduced_Marks			
▶	85			
	75			
	65			
	55			

Result 8 Result 9 × Result 10 Result 11 Result 12

Result Grid		Filter Rows:	Export:	Wrap Cell
	Double_Marks			
▶	180			
	160			
	140			
	120			

Result 8 Result 9 **Result 10** × Result 11 Result 12

Result Grid		Filter Rows:	Export:	Wrap Cell
	Half_Marks			
▶	45.0000			
	40.0000			
	35.0000			
	30.0000			

Result 8 Result 9 Result 10 **Result 11** × Result 12

Result Grid		Filter Rows:	Export:	Wrap Cell
	Remainder			
▶	0			
	0			
	0			
	0			

Result 8 Result 9 Result 10 Result 11 **Result 12** ×

Comparison Operators

```

CREATE DATABASE E;
USE E;
CREATE TABLE Student (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Marks INT,
    DOB DATE
);
INSERT INTO Student VALUES
(101, 'Manu', 90, '2005-05-10'),
(102, 'Varun', 80, '2004-08-15'),
(103, 'Vaibhav', 70, '2005-01-20'),
(104, 'Raghavendra', 60, '2004-12-05');
SELECT * FROM Student
WHERE Marks = 80;
SELECT * FROM Student
WHERE Marks > 70;
SELECT * FROM Student
WHERE Marks < 80;
SELECT * FROM Student
WHERE Marks >= 80;
SELECT * FROM Student
WHERE Marks <= 70;
SELECT * FROM Student
WHERE Marks <> 80;

```

Output

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Student_ID	Student_Name	Marks	DOB
▶	102	Varun	80	2004-08-15
Student 13 Student 14 Student 15 Student 16 Student 17 Student 18				

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Student_ID	Student_Name	Marks	DOB
▶	103	Vaibhav	70	2005-01-20
	104	Raghavendra	60	2004-12-05

Student 13
 Student 14
 Student 15 ×
 Student 16
 Student 17
 Student 18

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	102	Varun	80	2004-08-15

Student 13
 Student 14
 Student 15
 Student 16 ×
 Student 17
 Student 18

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	102	Varun	80	2004-08-15

Student 13
 Student 14 ×
 Student 15
 Student 16
 Student 17
 Student 18

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Student_ID	Student_Name	Marks	DOB
▶	103	Vaibhav	70	2005-01-20
	104	Raghavendra	60	2004-12-05

Student 13
 Student 14
 Student 15
 Student 16
 Student 17 ×
 Student 18

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Student_ID	Student_Name	Marks	DOB
101	Manu	90	2005-05-10
103	Vaibhav	70	2005-01-20
104	Raghavendra	60	2004-12-05

Student 13 Student 14 Student 15 Student 16 Student 17 Student 18 x

Logical Operators

```

CREATE DATABASE F;
USE F;
CREATE TABLE Student (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Marks INT,
    DOB DATE
);
INSERT INTO Student VALUES
(101, 'Manu', 90, '2005-05-10'),
(102, 'Varun', 80, '2004-08-15'),
(103, 'Vaibhav', 70, '2005-01-20'),
(104, 'Raghavendra', 60, '2004-12-05');
SELECT * FROM Student
WHERE Marks > 70 AND Marks < 90;
SELECT * FROM Student
WHERE Marks = 90 OR Marks = 60;
SELECT * FROM Student
WHERE NOT Marks = 80;

```

Output

Result Grid   Filter Rows: | Export:  | Wrap Cell Content

	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	104	Raghavendra	60	2004-12-05

Student 19 Student 20 × Student 21

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Student_ID	Student_Name	Marks	DOB
▶	102	Varun	80	2004-08-15

Student 19

×

Student 20

Student 21

Result Grid



 Filter Rows:

Export: 

Wrap Cell Content: ☐

	Student_ID	Student_Name	Marks	DOB
	102	Varun	80	2004-08-15

Student 19

×

Student 20

Student 21

SQL Operators – Special Operators and Set Operations

CREATE DATABASE G;

```

USE G;
CREATE TABLE Student (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Marks INT,
    DOB DATE
);
INSERT INTO Student VALUES
(101, 'Manu', 90, '2005-05-10'),
(102, 'Varun', 80, '2004-08-15'),
(103, 'Vaibhav', 70, '2005-01-20'),
(104, 'Raghavendra', 60, '2004-12-05');
SELECT * FROM Student
WHERE Marks BETWEEN 70 AND 90;
SELECT * FROM Student
WHERE Marks IN (60, 80, 90);
SELECT * FROM Student
WHERE Student_Name LIKE 'M%';
SELECT * FROM Student
WHERE Student_Name LIKE '%a%';
SELECT * FROM Student
WHERE DOB IS NULL;

```

Output

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	102	Varun	80	2004-08-15
	103	Vaibhav	70	2005-01-20

Student 22 × Student 23 Student 24 Student 25 Student 26

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	102	Varun	80	2004-08-15
	104	Raghavendra	60	2004-12-05

104

Student 22Student 23 × Student 24Student 25Student 26

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10

Student 22Student 23Student 24 × Student 25Student 26

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Student_ID	Student_Name	Marks	DOB
▶	101	Manu	90	2005-05-10
	102	Varun	80	2004-08-15
	103	Vaibhav	70	2005-01-20
	104	Raghavendra	60	2004-12-05

Student 22Student 23Student 24Student 25 × Student 26

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Student_ID	Student_Name	Marks	DOB
--	------------	--------------	-------	-----

Student 22Student 23Student 24Student 25Student 26 ×

Set Operations UNION

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search, along with a dropdown menu set to "Limit to 1000 rows". The SQL editor contains the following code:

```
1 • CREATE DATABASE H;  
2 • USE H;  
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Student_Name VARCHAR(50),  
6     Marks INT,  
7     DOB DATE  
8 );  
9 • INSERT INTO Student VALUES  
10 (101, 'Manu', 90, '2005-05-10'),  
11 (102, 'Varun', 80, '2004-08-15'),  
12 (103, 'Vaibhav', 70, '2005-01-20'),  
13 (104, 'Raghavendra', 60, '2004-12-05');  
14 • SELECT Student_Name FROM Student  
15 WHERE Marks >= 80  
16 UNION  
17 SELECT Student_Name FROM Student  
18 WHERE Marks >= 70;
```

Below the editor is the "Result Grid" section. It includes a "Filter Rows:" input field, an "Export:" button, and a "Wrap Cell Content:" checkbox. The result grid displays the output of the query:

Student_Name
Manu
Varun
Vaibhav

At the bottom, a tab labeled "Result 27" is visible.